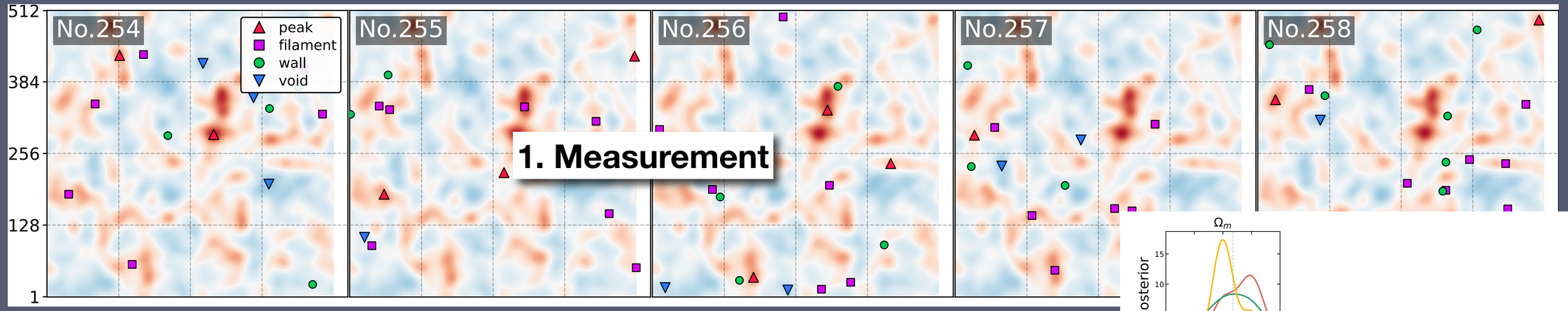
Abundance prediction of critical point & Betti numbers

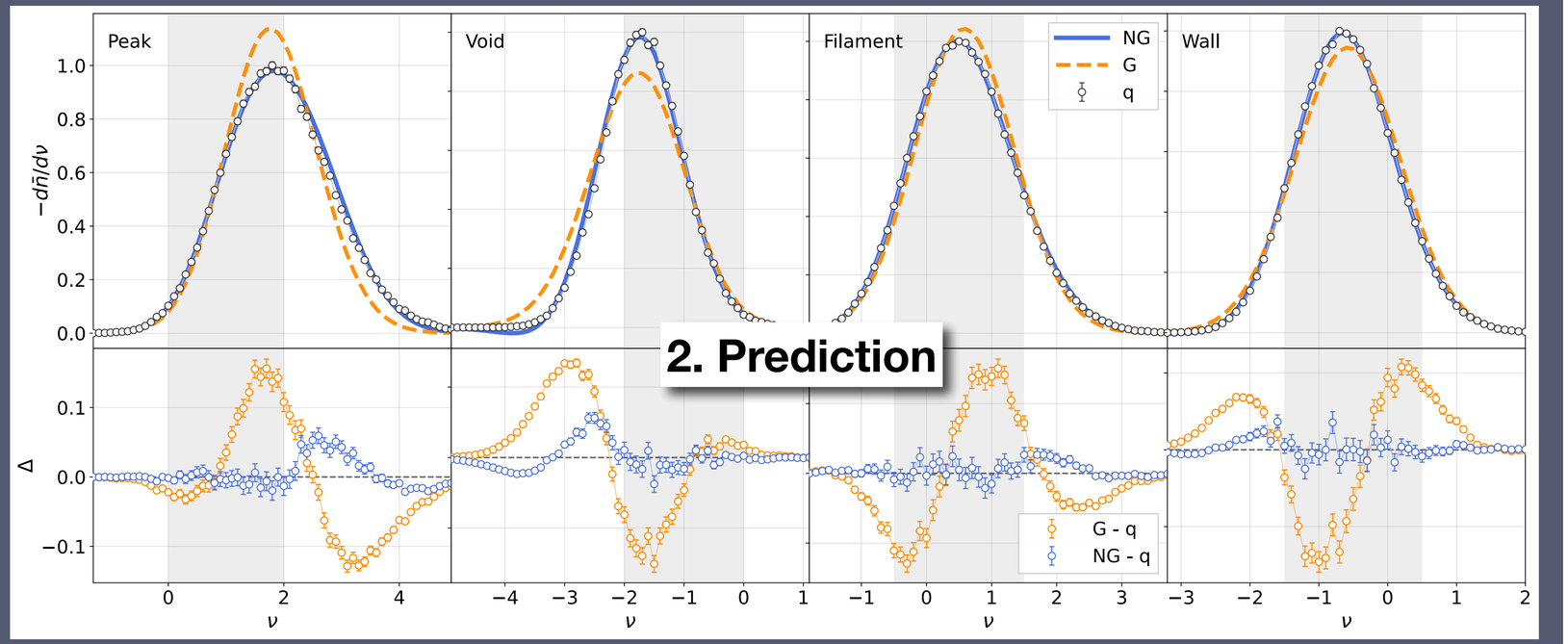
Yingqiu He (D1)

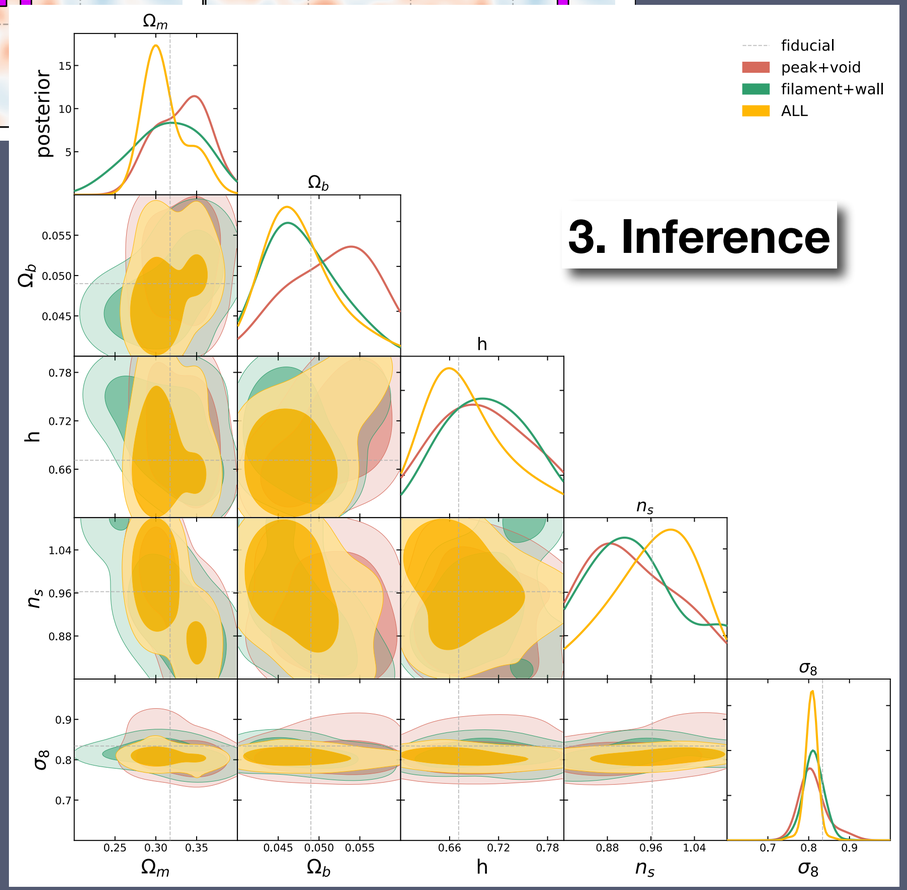
Cosmo-physics group internal talk, 2026.11.12

with substantial ChatGPT assistance

Previous study on leading-order correction







Review of critical-point statistics to σ^4

$$P(x) = \left[1 + \sigma\frac{S_3}{3!}H_3 + \sigma^2\left(\frac{S_4}{4!}H_4 + \frac{S_3^2}{2!(3!)^2}H_6\right) + \sigma^3\left(\frac{S_5}{5!}H_5 + \frac{S_3S_4}{3!4!}H_7 + \frac{S_3^3}{3!(3!)^3}H_9\right) + \sigma^4\left(\frac{S_6}{6!}H_6 + \left[\frac{S_3S_5}{3!5!} + \frac{S_4^2}{2!(4!)^2}\right]H_8 + \frac{S_3^2S_4}{2!(3!)^2 4!}H_{10} + \frac{S_3^4}{4!(3!)^4}H_{12}\right) + \cdots\right] P_{\rm G}(x)$$

$$\frac{d^n}{dx^n}e^{-x^2/2} = (-1)^n H_n(x) e^{-x^2/2}$$

$$X = \{f, \nabla f, H = \nabla\nabla f\} \quad \rightarrow \quad \{f, |\nabla f|^2, \mathrm{tr}\, H, \ldots\}$$

$$\bar{n}^{\rm pk} \equiv \langle n^{\rm pk}(\nu)\rangle = \langle \Theta(f/\sigma - \nu)\, \delta_{\rm D}^3(\nabla f)\, \Theta(\lambda_3) \left|\det H\right| \rangle$$

Review of critical-point statistics to σ^4

$$P(x) = \left[1 + \sigma\frac{S_3}{3!}H_3 + \sigma^2\left(\frac{S_4}{4!}H_4 + \frac{S_3^2}{2!(3!)^2}H_6\right) + \sigma^3\left(\frac{S_5}{5!}H_5 + \frac{S_3S_4}{3!4!}H_7 + \frac{S_3^3}{3!(3!)^3}H_9\right) + \sigma^4\left(\frac{S_6}{6!}H_6 + \left[\frac{S_3S_5}{3!5!} + \frac{S_4^2}{2!(4!)^2}\right]H_8 + \frac{S_3^2S_4}{2!(3!)^2 4!}H_{10} + \frac{S_3^4}{4!(3!)^4}H_{12}\right) + \cdots\right]P_{\mathrm{G}}(x)$$

$$\frac{d^n}{dx^n}e^{-x^2/2} = (-1)^n H_n(x)e^{-x^2/2}$$

$$\begin{aligned}\langle n^\star[f]\rangle = \langle n^\star[f]\rangle_{\mathrm{G}} &+ \frac{1}{3!}\int_{\boldsymbol{k}_{1\cdots3}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\mathcal{G}_3^\star \\ &+ \frac{1}{4!}\int_{\boldsymbol{k}_{1\cdots4}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\rangle_{\mathrm{c}}\,\mathcal{G}_4^\star + \frac{1}{2(3!)^2}\int_{\boldsymbol{k}_{1\cdots6}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\langle\tilde{f}_4\tilde{f}_5\tilde{f}_6\rangle_{\mathrm{c}}\,\mathcal{G}_6^\star \\ &+ \frac{1}{5!}\int_{\boldsymbol{k}_{1\cdots5}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\tilde{f}_5\rangle_{\mathrm{c}}\,\mathcal{G}_5^\star + \frac{1}{3!4!}\int_{\boldsymbol{k}_{1\cdots7}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\rangle_{\mathrm{c}}\,\langle\tilde{f}_5\tilde{f}_6\tilde{f}_7\rangle_{\mathrm{c}}\,\mathcal{G}_7^\star + \frac{1}{3!(3!)^3}\int_{\boldsymbol{k}_{1\cdots9}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\langle\tilde{f}_4\tilde{f}_5\tilde{f}_6\rangle_{\mathrm{c}}\,\langle\tilde{f}_7\tilde{f}_8\tilde{f}_9\rangle_{\mathrm{c}}\,\mathcal{G}_9^\star \\ &+ \frac{1}{6!}\int_{\boldsymbol{k}_{1\cdots6}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\tilde{f}_5\tilde{f}_6\rangle_{\mathrm{c}}\,\mathcal{G}_6^\star + \frac{1}{3!5!}\int_{\boldsymbol{k}_{1\cdots8}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\langle\tilde{f}_4\tilde{f}_5\tilde{f}_6\tilde{f}_7\tilde{f}_8\rangle_{\mathrm{c}}\,\mathcal{G}_8^\star + \frac{1}{2(4!)^2}\int_{\boldsymbol{k}_{1\cdots8}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\rangle_{\mathrm{c}}\,\langle\tilde{f}_5\tilde{f}_6\tilde{f}_7\tilde{f}_8\rangle_{\mathrm{c}}\,\mathcal{G}_8^\star \\ &+ \frac{1}{2(3!)^2 4!}\int_{\boldsymbol{k}_{1\cdots10}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\langle\tilde{f}_4\tilde{f}_5\tilde{f}_6\rangle_{\mathrm{c}}\,\langle\tilde{f}_7\tilde{f}_8\tilde{f}_9\tilde{f}_{10}\rangle_{\mathrm{c}}\,\mathcal{G}_{10}^\star + \frac{1}{4!(3!)^4}\int_{\boldsymbol{k}_{1\cdots12}}\langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\mathrm{c}}\,\langle\tilde{f}_4\tilde{f}_5\tilde{f}_6\rangle_{\mathrm{c}}\,\langle\tilde{f}_7\tilde{f}_8\tilde{f}_9\rangle_{\mathrm{c}}\,\langle\tilde{f}_{10}\tilde{f}_{11}\tilde{f}_{12}\rangle_{\mathrm{c}}\,\mathcal{G}^\star\end{aligned}$$

$\mathcal{G}_n^\star$: Gaussian Fourier response

Outline

- Previous approach (May, GPT-5)
  - Computed $\frac{1}{4!}\int_{\boldsymbol{k}_{1\cdots4}} \langle \tilde{f}_1 \tilde{f}_2 \tilde{f}_3 \tilde{f}_4 \rangle_{\mathrm{c}}\, \mathcal{G}_4^{\star}$ need traceless symmetric rank-8 tensor
  - Problem: extension to high order requires complicated tensor contractions
- Improved approach (Sept. GPT-6)
  - Avoid explicit high rank tensor construction and contractions
  - Direct Fourier coefficients from polynomials

- Can we learn Betti numbers from $P(k)$? Covering diverse $P(k)$ is difficult.

1. Polynomial basis from Gaussian PDF

Hermite polynomials: $\frac{d^n}{dx^n}e^{-x^2/2} = (-1)^n H_n(x) e^{-x^2/2}$

* to calculate Gaussian derivatives, don't need derivative operators

Non-Gaussian PDF cumulant expansion: $P_{\rm NG} = P_{\rm G}\left[1 + \frac{S_3}{3!}H_3 + \frac{S_4}{4!}H_4 + \cdots\right]$

For multi-variables: $(-1)^n \partial_{a_1} \cdots \partial_{a_n} P_{\rm G}(X) = H_{a_1 \cdots a_n}(X) P_{\rm G}(X)$

Matsubara (1994), Gay et al. (2011), etc.

2. Rotationally invariant polynomial basis for 2D

$$X = (\alpha, \eta_x, \eta_y, \zeta_{xx}, \zeta_{xy}, \zeta_{yy}) \;\rightarrow\; Y = (\alpha, \eta^2, \mathrm{tr}\,\zeta, \mathrm{tr}\,\hat{\zeta}^2)$$

Eg. $P_{\mathrm{G}}(\eta^2) \sim \exp(-\eta^2) \quad \eta^2 \geq 0$

Lagerre polynomial: $L_n(x) = \dfrac{e^x}{n!}\dfrac{d^n}{dx^n}(e^{-x}x^n)$ for η^2

Rotational-invariant Gaussian polynomial basis for 2D:

$$B_{ijkl}(Y) \sim \underbrace{H_{ij}(\alpha, \mathrm{tr}\,\zeta)}_{\text{Gassian scalar}} \times \underbrace{L_{kl}(\eta^2, \mathrm{tr}\,\hat{\zeta}^2)}_{\text{square of vectors}}$$

Gay et al. (2011), Lazeyras et al. (2016), etc.

3. Gaussian response and Fourier coefficient

Define $G^{\star}_{ijkl}(\nu) \equiv \langle n^{\star}(\nu)\, B_{ijkl}(Y)\rangle_{\rm G}$

$$B_{ijkl}(Y) \sim H_{ij}(\alpha, \mathrm{tr}\,\zeta) \times L_{kl}(\eta^2, \mathrm{tr}\,\hat{\xi}^2)$$

* as Gaussian expectation of $n^{\star}$ for corresponding polynomial (or projection)

Define $\mathscr{G}^{\star}_{n}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n; \nu) \equiv (-1)^n \int dX\, n^{\star}(\nu)\, \widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n P_{\rm G}(X)$

* as Gaussian Fourier response kernel

$$\widehat{\mathscr{D}}_a = \frac{1}{\sigma_0}\partial_\alpha + \frac{ik_{ai}}{\sigma_1}\partial_{\eta_i} - \frac{k_{ai}k_{aj}}{\sigma_2}\partial_{\zeta_{ij}}$$

$$\mathscr{G}^{\star}_{n} = \sum_{i+j+2k+2l=n} C^{(n)}_{ijkl}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)\, G^{\star}_{ijkl}(\nu)\,.$$

Fourier coefficient

Lazeyras et al. (2016), Matsubara (2020), etc.

Recap:

$$\mathcal{G}_n^\star(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n;\nu) = (-1)^n \int dY\, n^\star(\nu)\, \widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n P_{\rm G}(Y)$$

$$= \sum_{i+j+2k+2l=n} C^{(n)}_{ijkl}(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n) \left\langle n^\star(\nu)\, B_{ijkl}(Y) \right\rangle_{\rm G}$$

Previous: deal with complicated derivative tensors construction

$$\langle n^\star[f]\rangle = \langle n^\star[f]\rangle_{\rm G} + \frac{1}{3!}\int_{\boldsymbol{k}_{1\cdots3}} \langle \tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\rm c}\ \mathcal{G}_3^\star$$

Recap:

$$\mathcal{G}_n^\star(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n;\nu) = (-1)^n \int dY\, n^\star(\nu)\, \widehat{\mathcal{D}}_1 \cdots \widehat{\mathcal{D}}_n P_{\rm G}(Y)$$

$$= \sum_{i+j+2k+2l=n} C^{(n)}_{ijkl}(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n) \left\langle n^\star(\nu)\, B_{ijkl}(Y) \right\rangle_{\rm G}$$

Previous: deal with complicated derivative tensors construction

$$\langle n^\star[f]\rangle = \langle n^\star[f]\rangle_{\rm G} + \frac{1}{3!}\int_{\boldsymbol{k}_{1\cdots3}} \langle \tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle_{\rm c}\, \mathcal{G}_3^\star$$

4. Fourier coefficient extraction (GPT-provided)

$$\Pi_{\rm inv} R_n(X) = \sum_\lambda C^{(n)}_\lambda B_\lambda(Y), \quad R_n(X) \equiv \frac{(-1)^n}{P_{\rm G}(X)} \widehat{\mathcal{D}}_1 \cdots \widehat{\mathcal{D}}_n P_{\rm G}(X) \quad \lambda,\rho = (i,j,k,l)$$

$$\langle B_\lambda B_\rho\rangle_G = h_\lambda \delta_{\lambda\rho} \text{ (orthogonal)} \quad \rightarrow \quad \langle \widehat{\mathcal{D}}_1 \cdots \widehat{\mathcal{D}}_n B_\lambda\rangle_{\rm G} = \langle B_\lambda R_n\rangle_{\rm G} = h_\lambda C^{(n)}_\lambda$$

Eg. $B_{0010} = \eta^2 - 1$, $\widehat{\mathcal{D}} = \frac{i}{\sigma_1}\boldsymbol{k}\cdot\nabla_\eta \rightarrow \widehat{\mathcal{D}}_2\widehat{\mathcal{D}}_1 B_{0010} = -\frac{2}{\sigma_1^2}\boldsymbol{k}_1\cdot\boldsymbol{k}_2 \rightarrow C^{(2)}_{0010}$

$$C_\lambda^{(n)} = \frac{1}{h_\lambda}\left\langle \widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n B_\lambda \right\rangle_{\mathrm{G}} = \text{constant}$$

$$\downarrow \qquad \underbrace{\widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n}_{n \text{ Fourier modes}} \iff \underbrace{[t_1 t_2 \cdots t_n]}_{\text{once for every mode}}$$

$$C_{ijkl}^{(n)} = [t_1 \cdots t_n] \frac{\alpha(\boldsymbol{t})^i}{i!} \frac{J_1(\boldsymbol{t})^j}{j!} \frac{\eta^2(\boldsymbol{t})^k}{k!} \frac{J_2(\boldsymbol{t})^l}{l!}, \qquad i + j + 2k + 2l = n$$

$$\alpha(\boldsymbol{t}) = \frac{1}{\sigma_0} \sum_a t_a,$$

$$J_1(\mathbf{t}) = \frac{1}{\sigma_2} \sum_a t_a k_a^2,$$

$$\eta^2(\boldsymbol{t}) = -\frac{1}{\sigma_1^2} \sum_{a,b} t_a t_b (\boldsymbol{k}_a \cdot \boldsymbol{k}_b),$$

$$J_2(\boldsymbol{t}) = \frac{2}{\sigma_2^2} \sum_{a,b} t_a t_b \left[(\boldsymbol{k}_a \cdot \boldsymbol{k}_b)^2 - \frac{k_a^2 k_b^2}{2} \right].$$

$$C_\lambda^{(n)} = \frac{1}{h_\lambda}\left\langle \widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n B_\lambda \right\rangle_{\mathrm{G}} = \text{constant}$$

intuitively

$$F(x) = x^2, \quad F(x + t_1 + t_2) = \cdots + 2t_1 t_2 + \cdots$$

$$F''(x) = \frac{\partial^2 F}{\partial t_1 \partial t_2}\bigg|_{\mathbf{t}=0} = [t_1 t_2] F(x + t_1 + t_2)$$

$$\downarrow \qquad \underbrace{\widehat{\mathscr{D}}_1 \cdots \widehat{\mathscr{D}}_n}_{n \text{ Fourier modes}} \iff \underbrace{[t_1 t_2 \cdots t_n]}_{\text{once for every mode}}$$

$$C_{ijkl}^{(n)} = [t_1 \cdots t_n] \frac{\alpha(\boldsymbol{t})^i}{i!} \frac{J_1(\boldsymbol{t})^j}{j!} \frac{\eta^2(\boldsymbol{t})^k}{k!} \frac{J_2(\boldsymbol{t})^l}{l!}, \qquad i + j + 2k + 2l = n$$

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Key advantage:

become ordinary polynomial expansions

$$\mathscr{G}_n^\star = \sum_\lambda C_\lambda^{(n)} G_\lambda^\star, \quad \bar{n}^\star = \sum_n \int dk_n \langle \tilde{f}_n \rangle \mathscr{G}_n^\star$$

$f = g + 0.1g^2 + 0.02g^3$

$./\bar{n}_{\mathrm{tot}} \rightarrow$

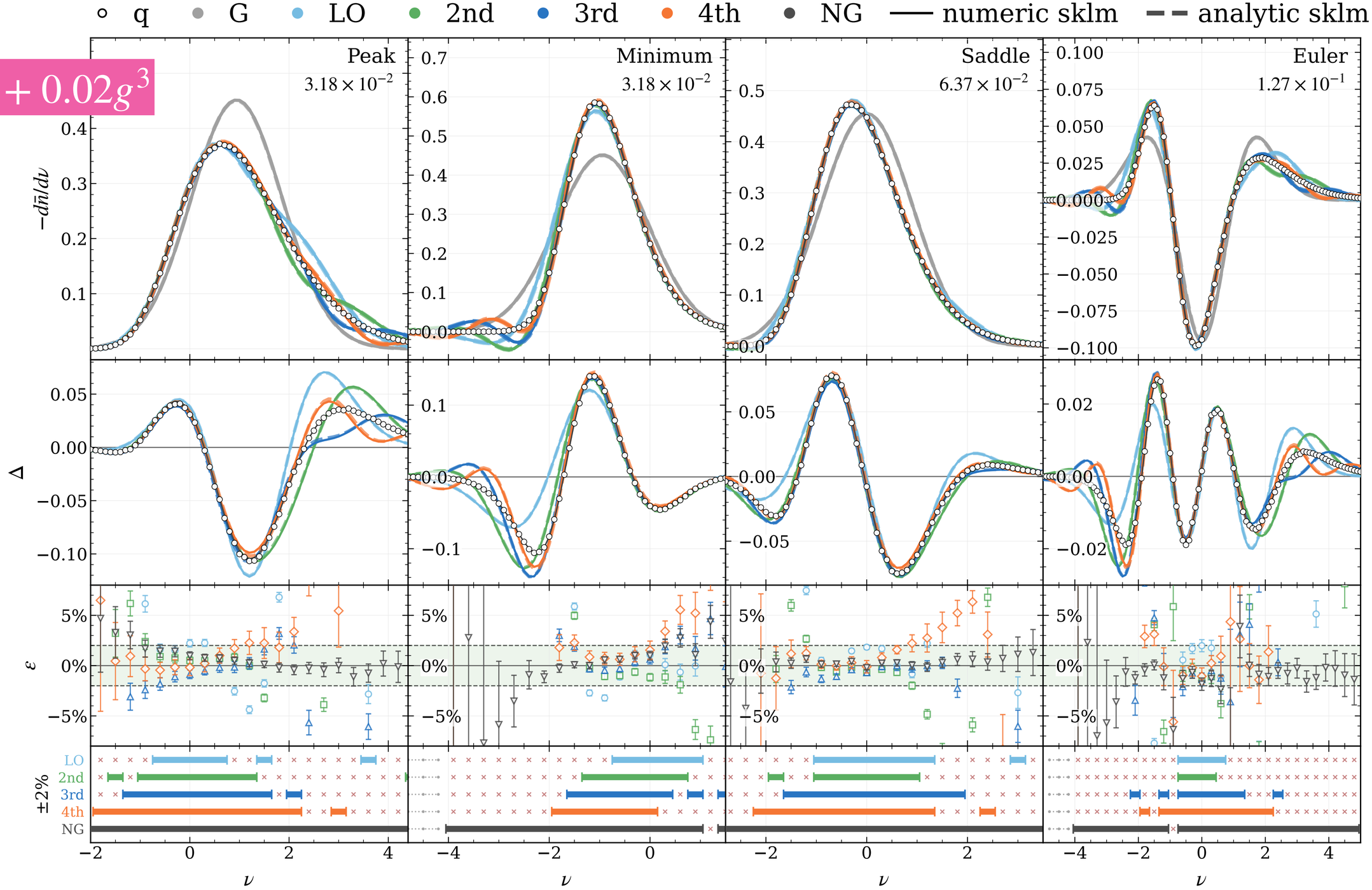
$. + \frac{d\bar{n}}{d\nu}|_{\mathrm{G}} \rightarrow$

$\Delta\nu = 0.3$

$\pm 2\,\mathrm{SEM}$

$\mu/q - 1 \rightarrow$

$2\,\%\,|q| \rightarrow$



$$f = g + 0.1g^2 + 0.02g^3$$

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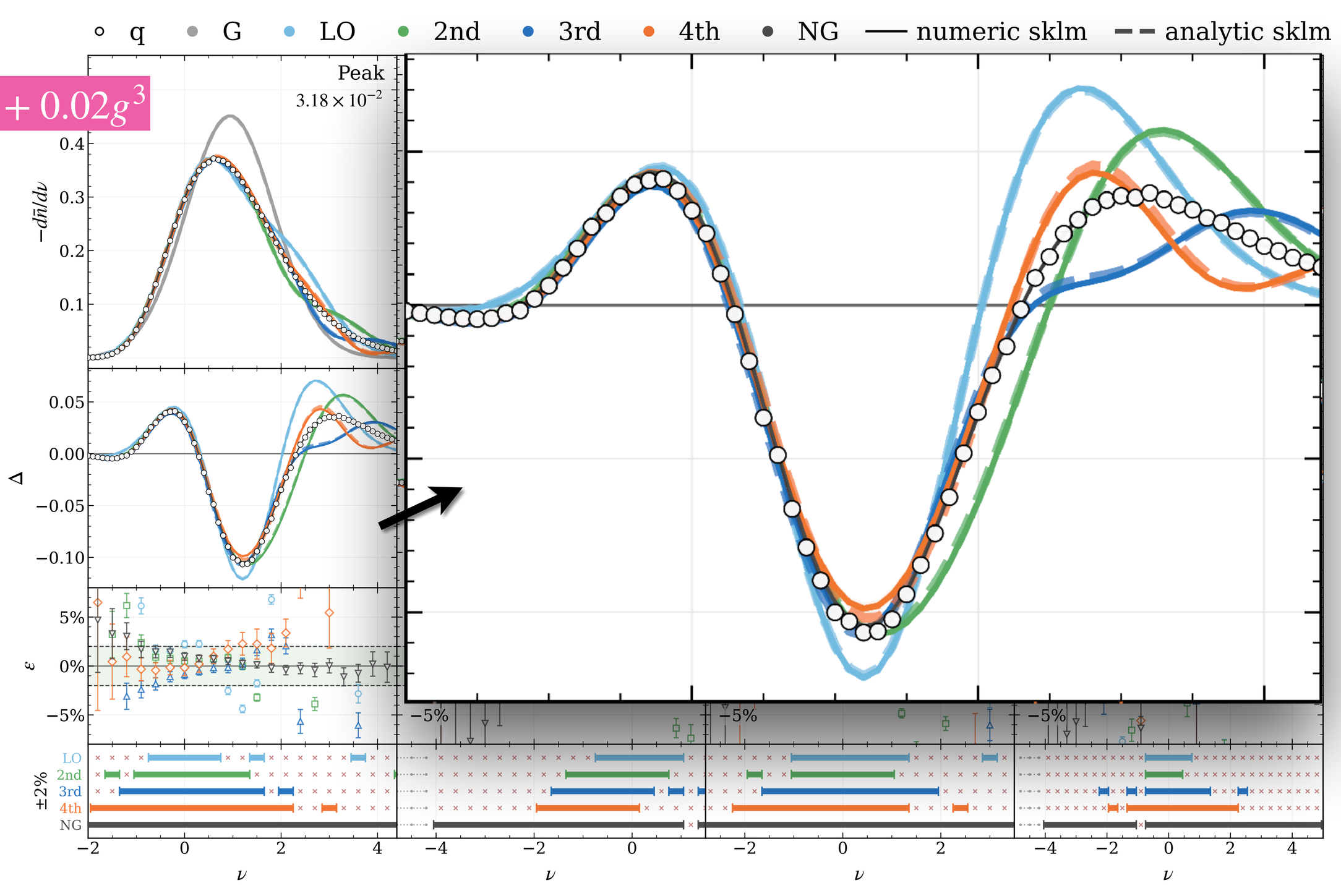
$. + \frac{d\bar{n}}{d\nu}\big|_{\mathrm{G}} \rightarrow$

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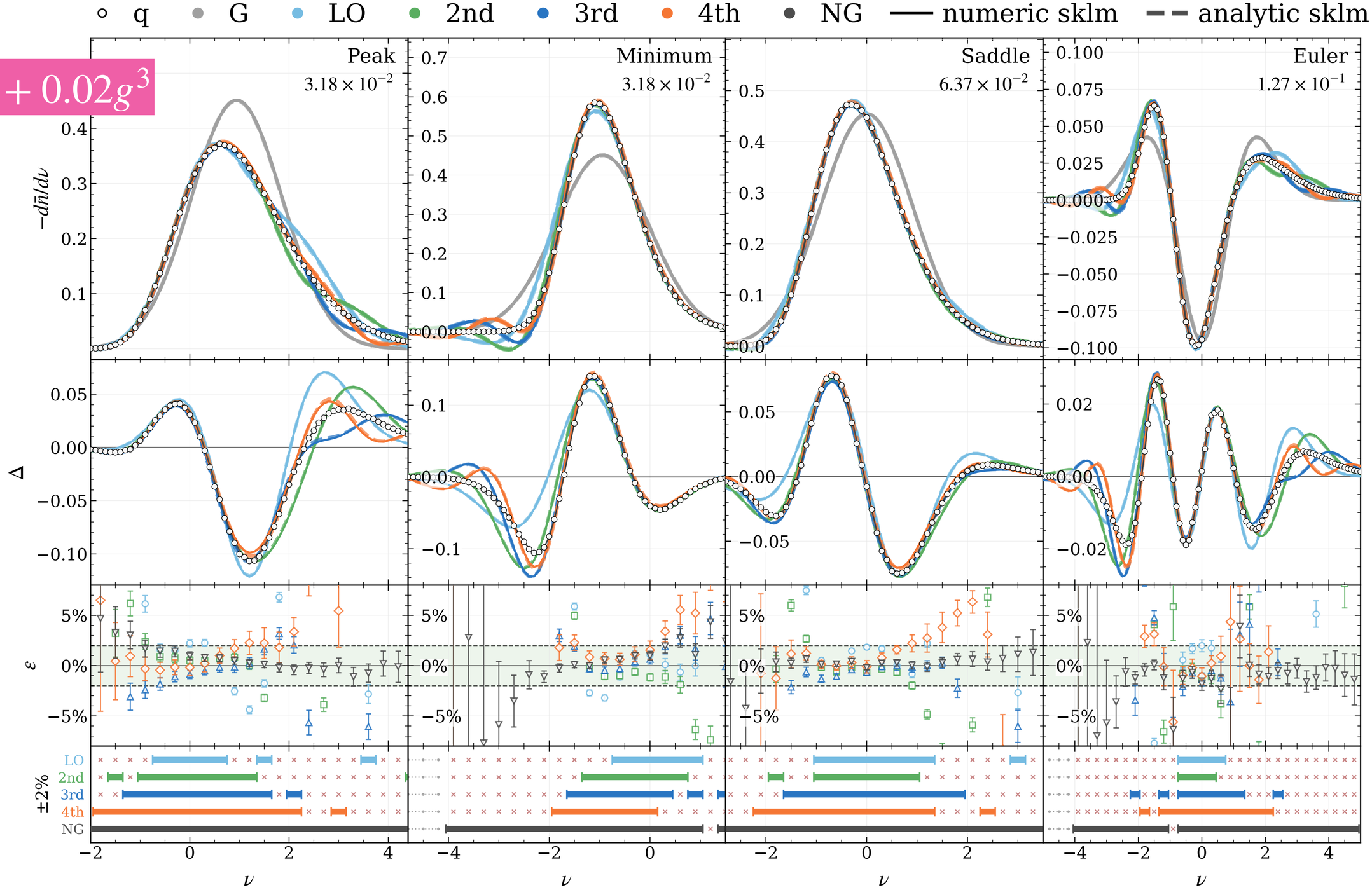
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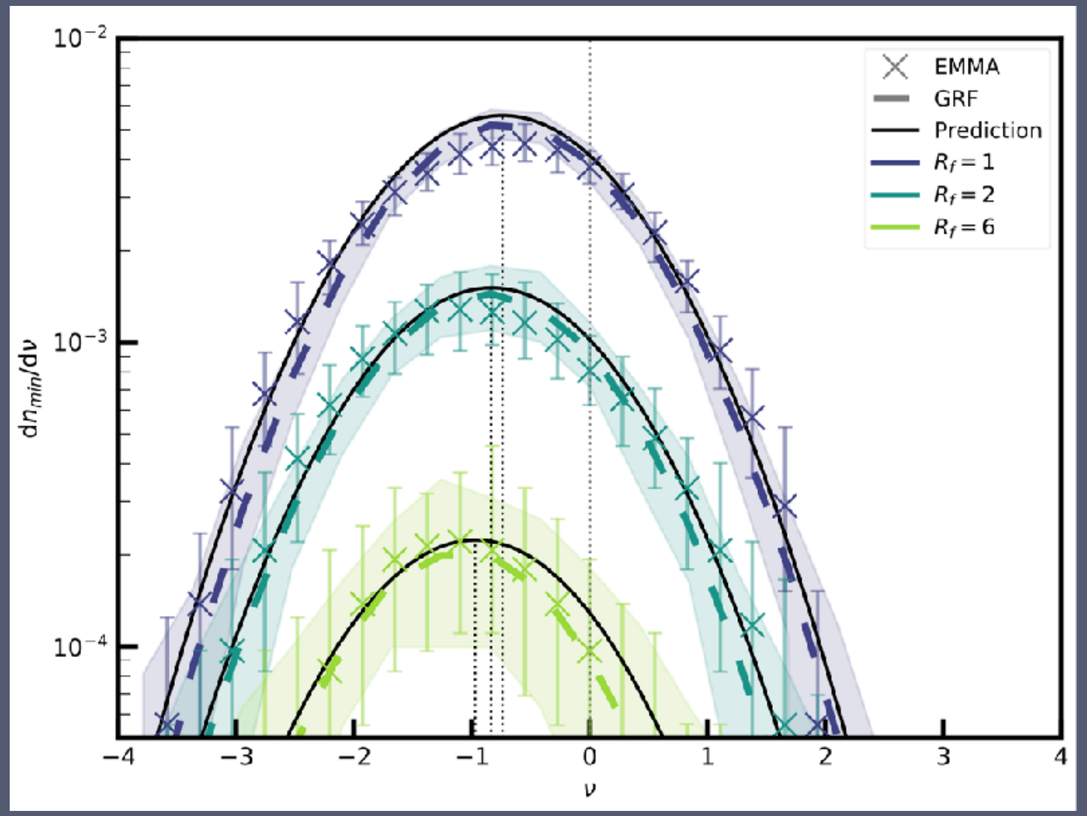
$\mu/q - 1 \rightarrow$

$2\,\%\,|q| \rightarrow$



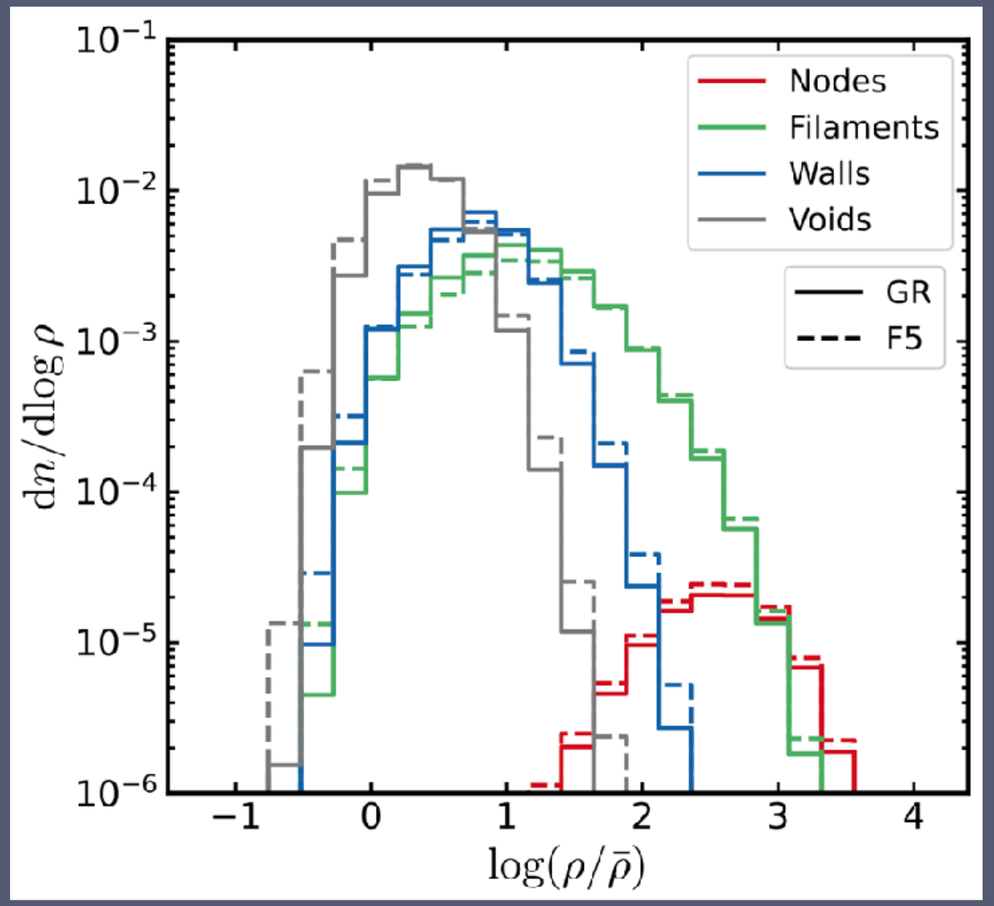
Application to other cosmology

21cm maps of cosmic reionisation



Thélie et al. (2023), Fig. 13

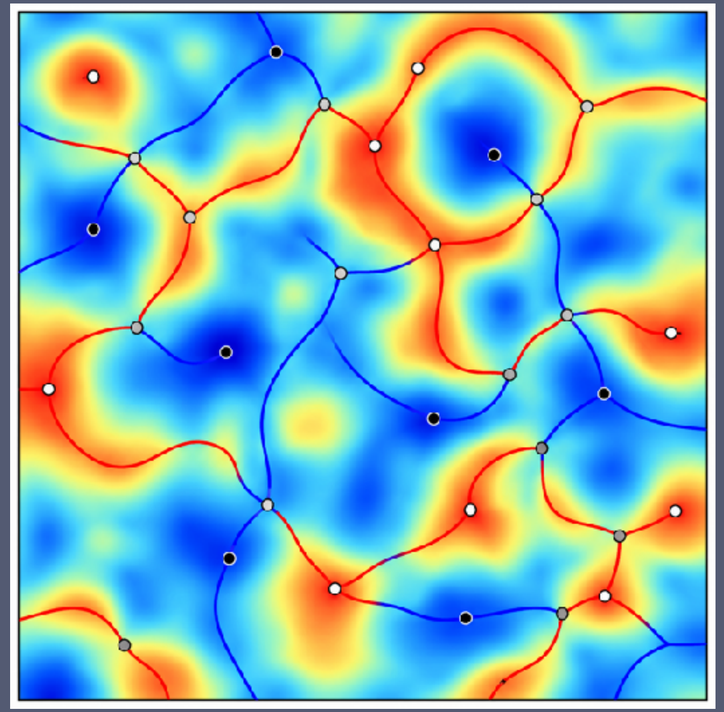
Modify gravity, fifth force simulation



Armijo & Da Costa (2025), Fig. 2

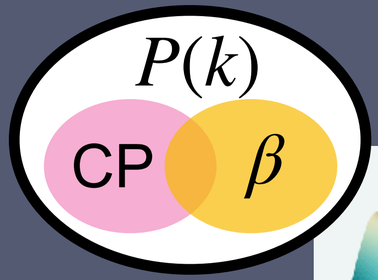
Extensions & limitations

- Given the required polyspectra, the framework can compute even higher order perturbative corrections to local geometric observables
- 4th order predictions agree well with weakly non-Gaussian regimes, but deviations still exist in highly non-Gaussian regions
- Extensions to other local geometry statistics:
  - Minkowski functionals (area, contour length, Euler)
  - Stiff ridges and valleys (curve length)
- ***Why Betti numbers cannot?*** It's topology feature.



Generated by ChatGPT

Betti numbers in 2D



β_0 : connected components $= N_{\text{peak}} - N_{\text{saddle,merge}}$

β_1 : independent loops $= N_{\text{saddle,loop}} - N_{\text{min}}$

* not consider boundary

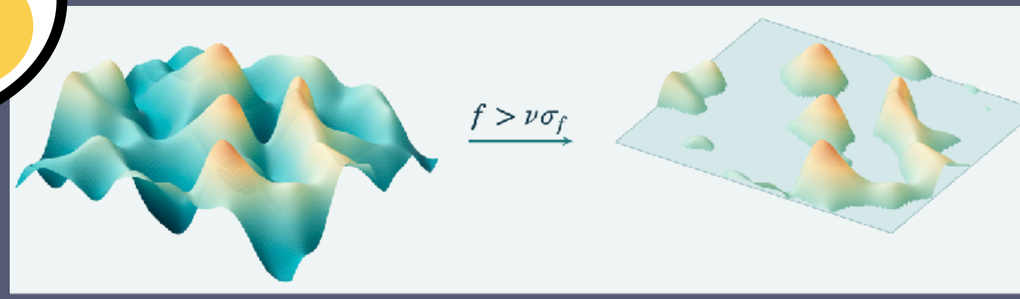
$$\chi = \beta_0 - \beta_1$$

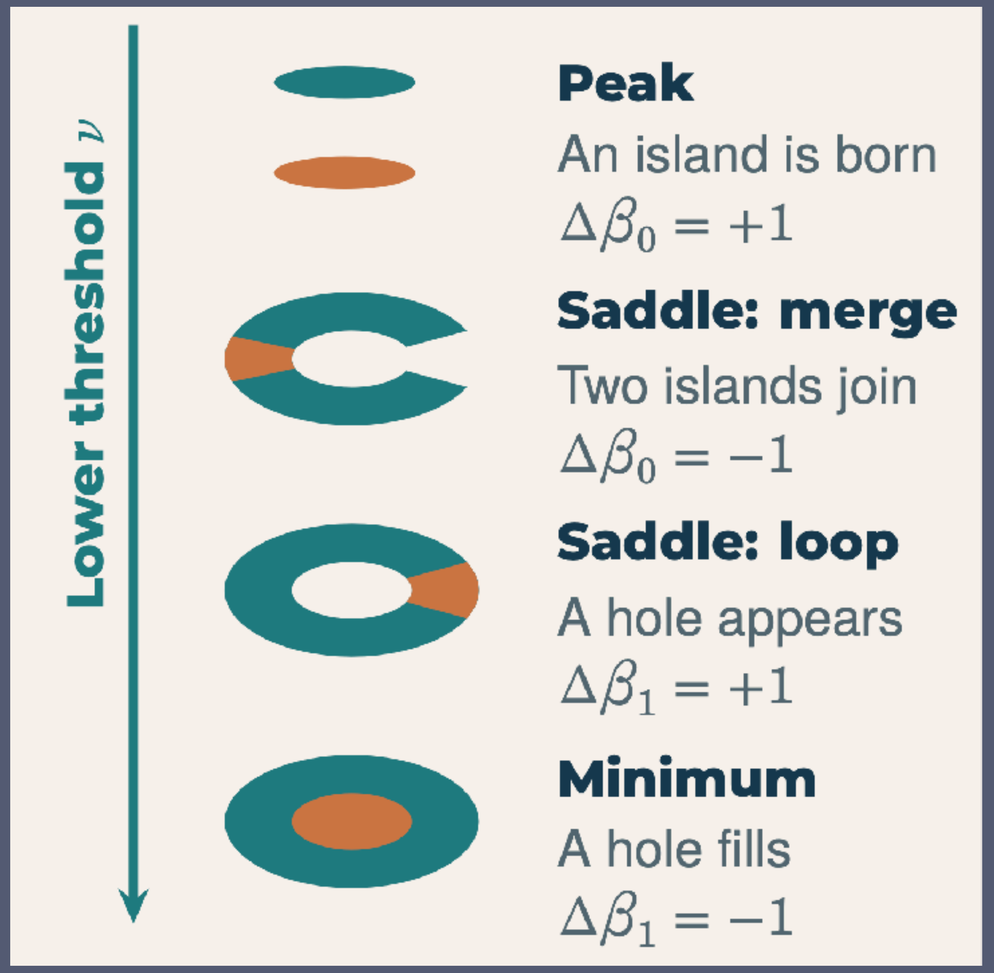
Depend on global connectivity,

cannot obtain from local one-point statistics

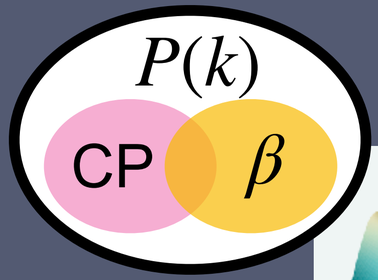
For Gaussian, $P(k)$ determine statistical distribution

including expected Betti numbers





Betti numbers in 2D



β_0 : connected components $= N_{\text{peak}} - N_{\text{saddle,merge}}$

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* not consider boundary

$\chi = \beta_0 - \beta_1$

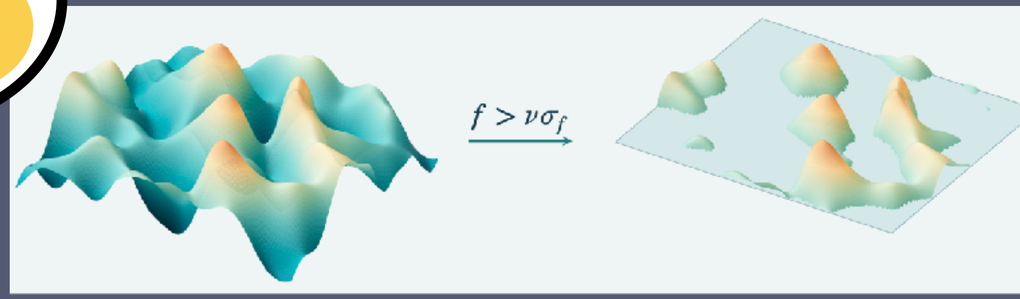
Depend on global connectivity,

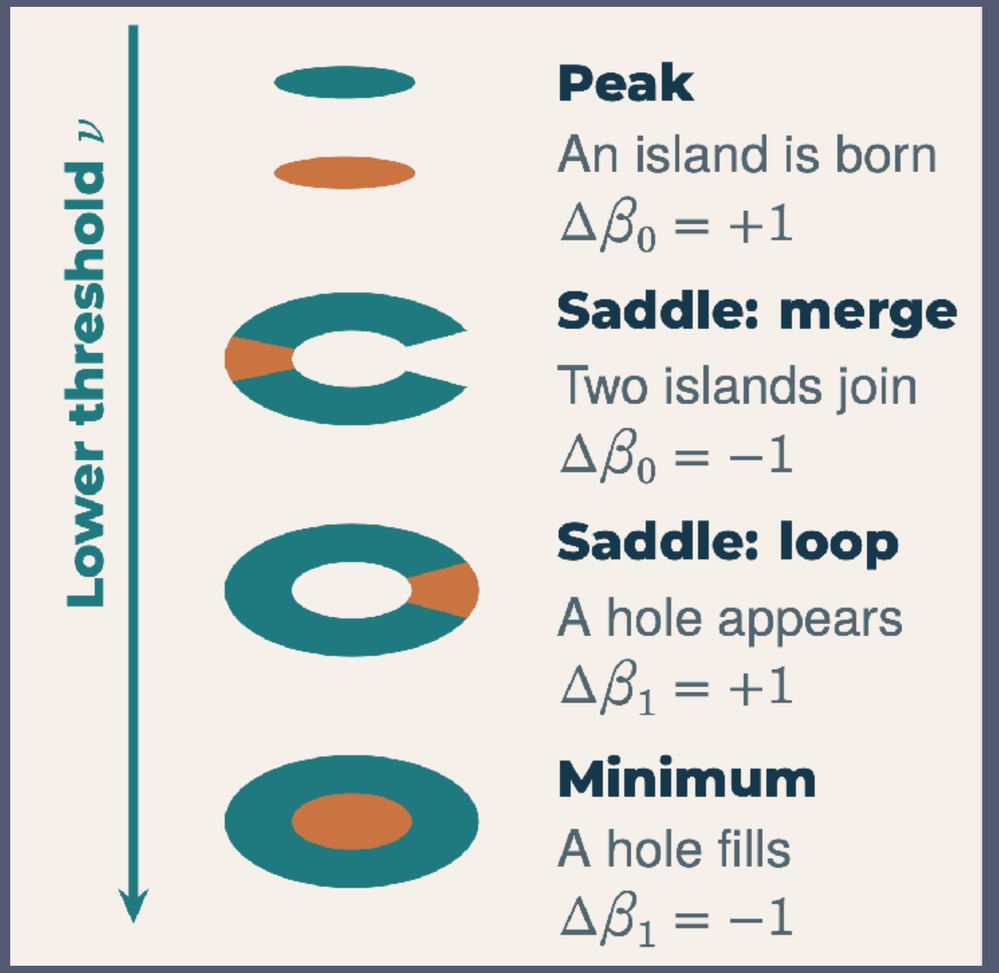
cannot obtain from local one-point statistics

For Gaussian, $P(k)$ determine statistical distribution

including expected Betti numbers

$$P(k) \xrightarrow{\text{Neural network}} \{\langle\beta_0(\nu)\rangle, \langle\beta_1(\nu)\rangle\}$$





Deep operator network

* MLP: multilayer perceptron

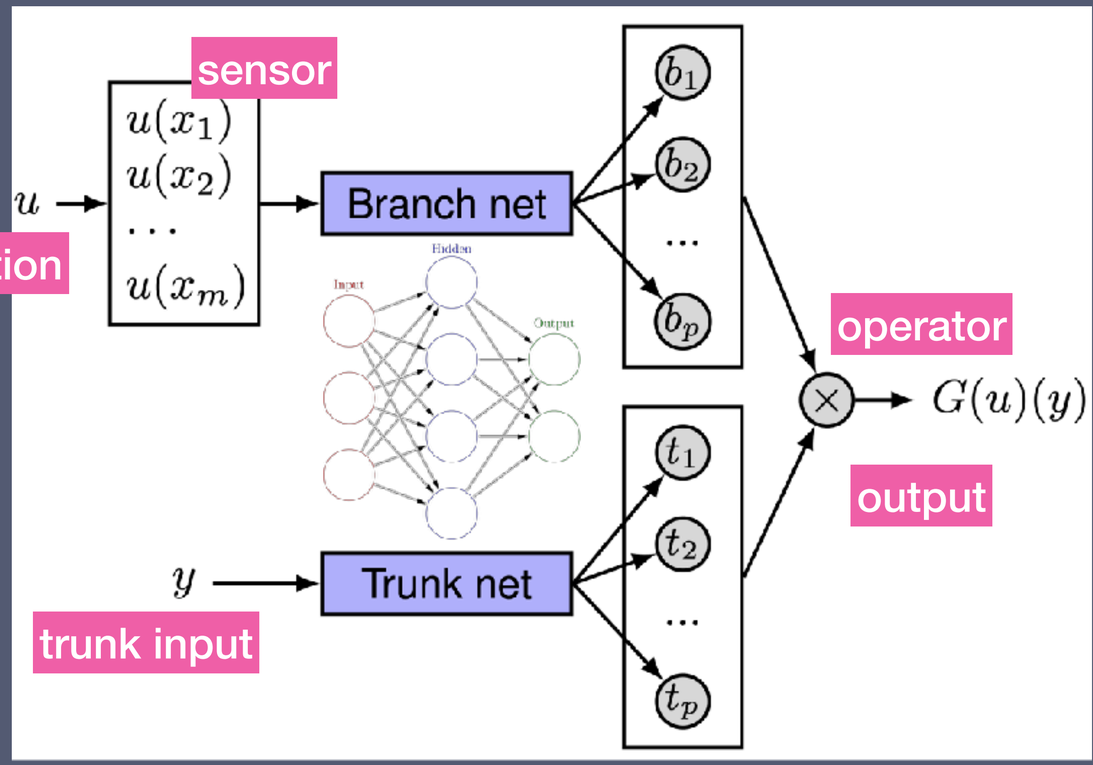
$$\underbrace{(P(k_1), \ldots, P(k_{256}))}_{\text{Branch input}} \xrightarrow{\text{Branch MLP}} (b_1, \ldots, b_p)$$

$$\underbrace{\nu}_{\text{Trunk input}} \xrightarrow{\text{Trunk MLP}} (t_1(\nu), \ldots, t_p(\nu))$$

Output: $\beta_i(\nu; P) \approx \sum_{a=1}^{p} b_{ia}[P]\ t_{ia}(\nu)$

$P(k)$-dependent coefficients

ν-dependent basis functions



Lu Lu et.al (2020), Fig 1

Training data

Gaussian process (GP):

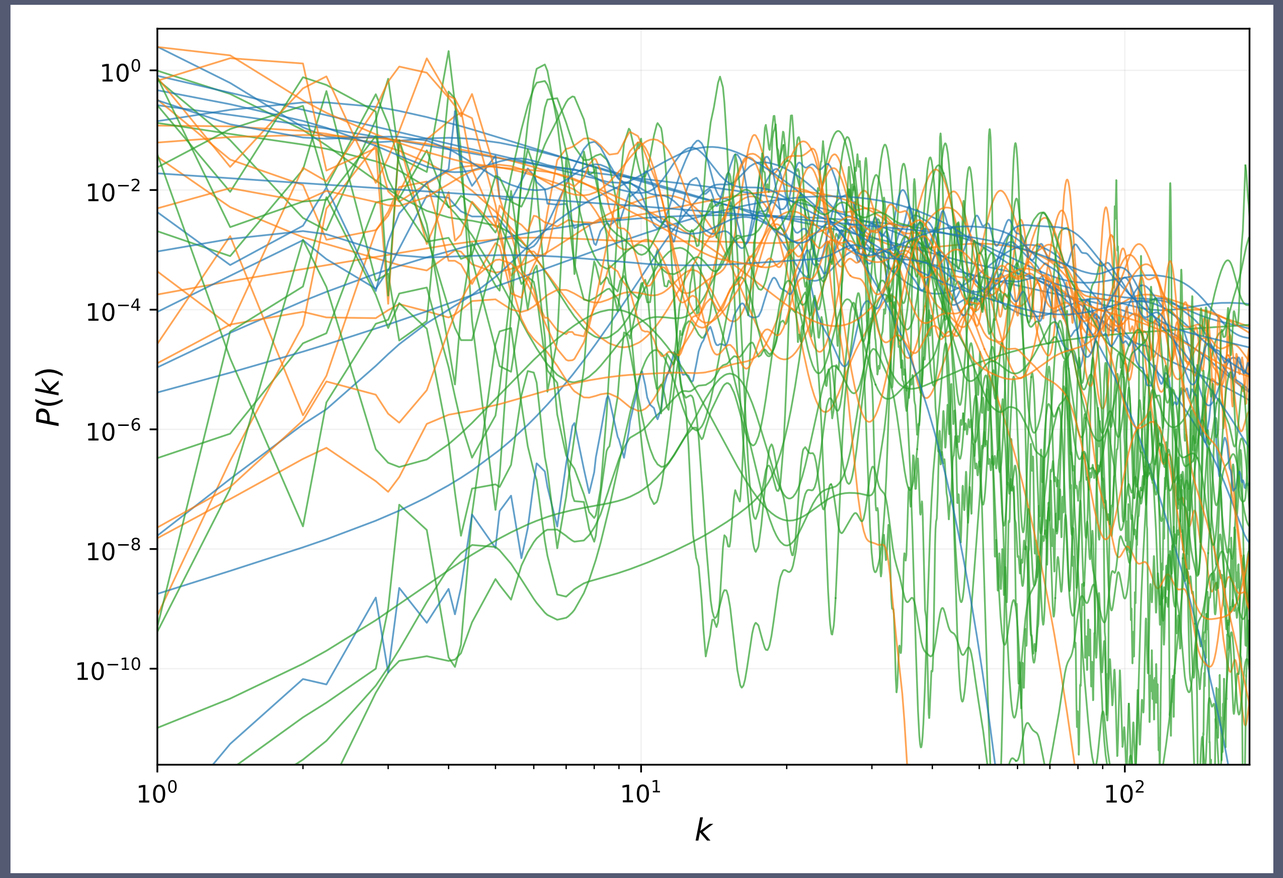
distribution over functions, characterized by

$$P(k) \sim \mathrm{GP}(m(k), K(k, k'))$$

mean function & covariance kernel

to controls the correlations

and smoothness of sampled functions

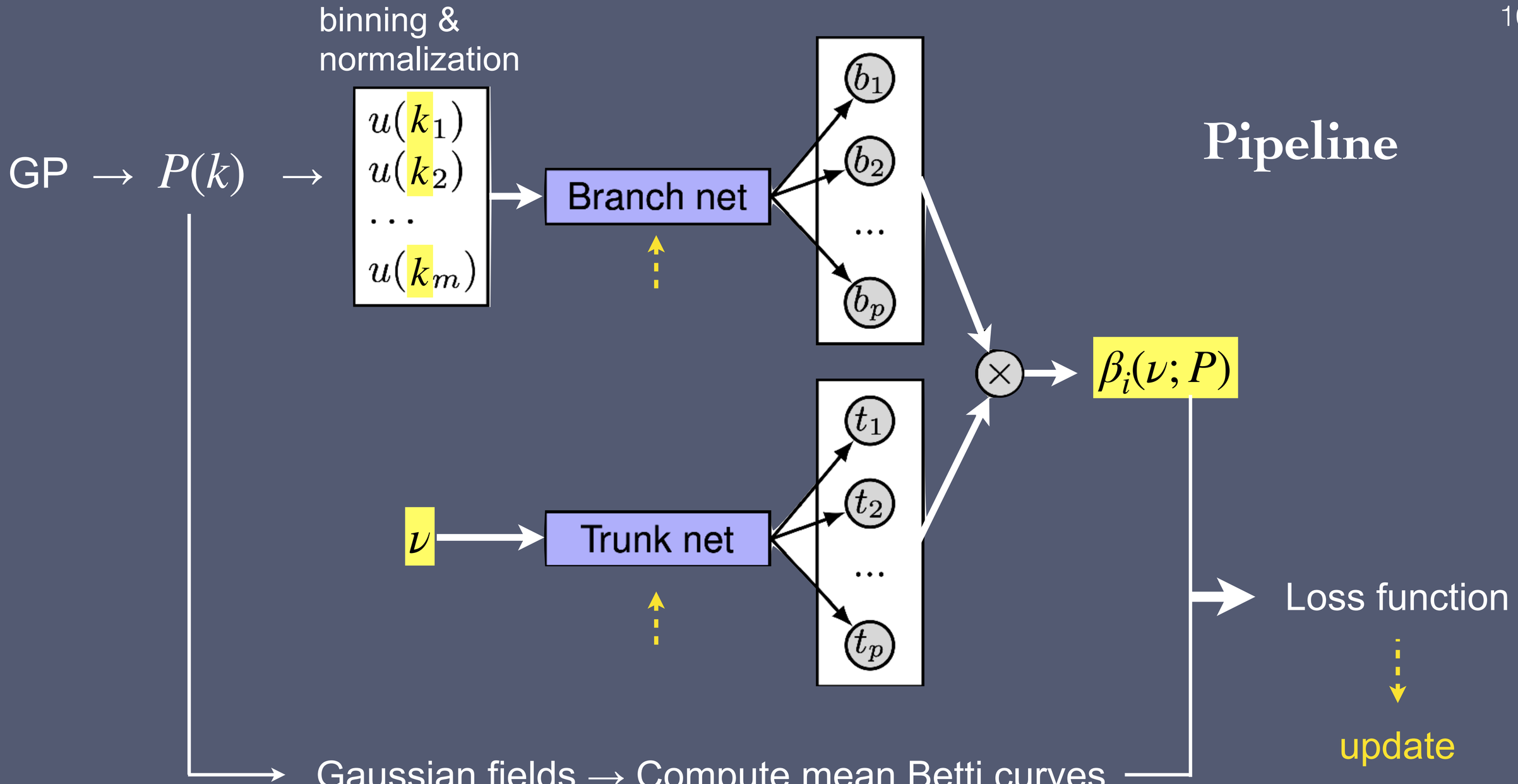


Input power spectra

GP → sample $P(k)$ → generate Gaussian-field realizations

→ compute and average Betti curves → input to neutral networks

Pipeline


binning &
normalization
GP → $P(k)$ →
$u(k_1)$
$u(k_2)$
...
$u(k_m)$
Branch net
b_1
b_2
...
b_p
ν
Trunk net
t_1
t_2
...
t_p
×
$\beta_i(\nu; P)$
Loss function
update
Gaussian fields → Compute mean Betti curves

Test in cosmology templates

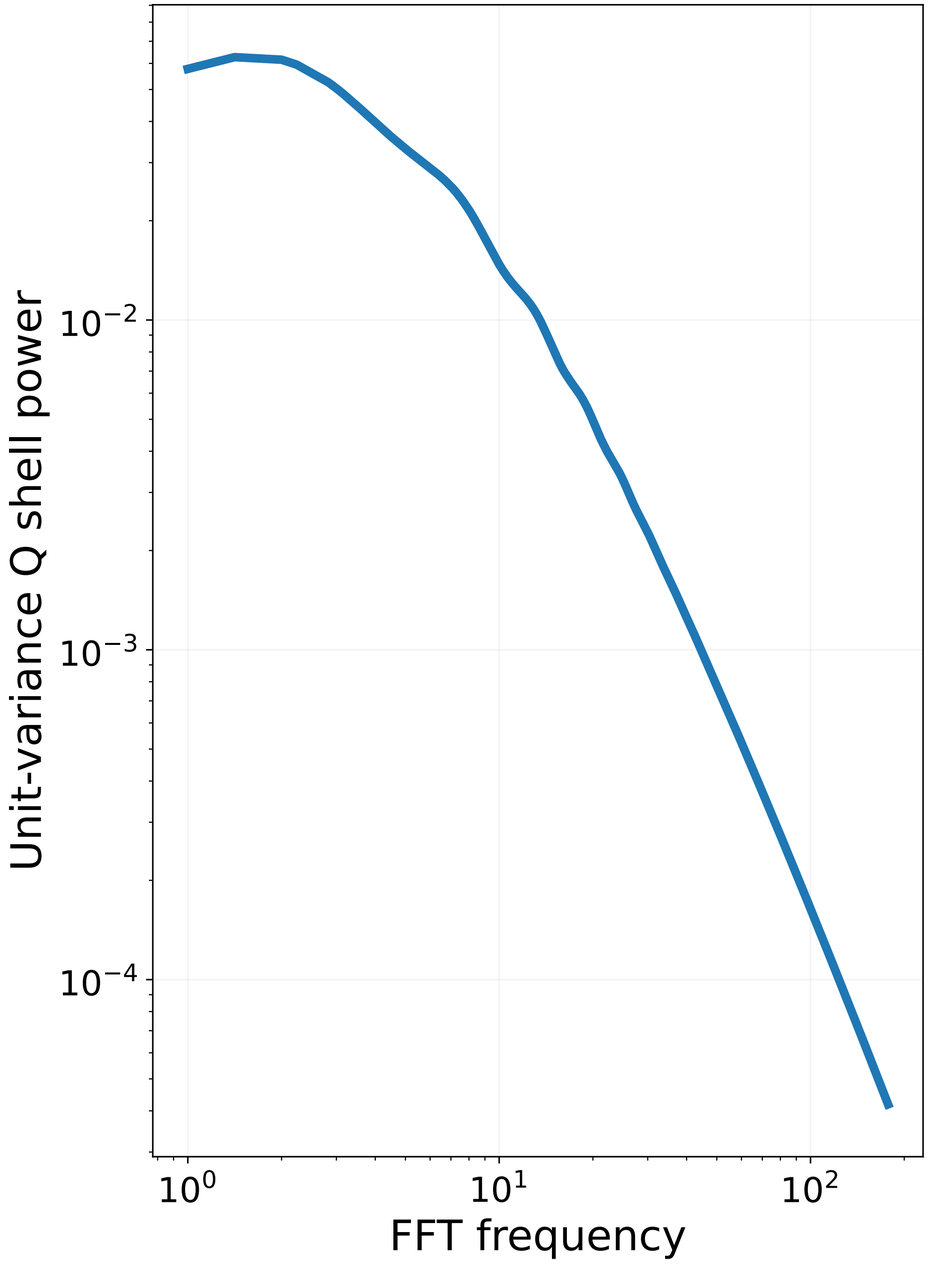
BAO: $Ak^{n_s}[(1-f_b)T_c(k)+f_bT_b(k)]^2$

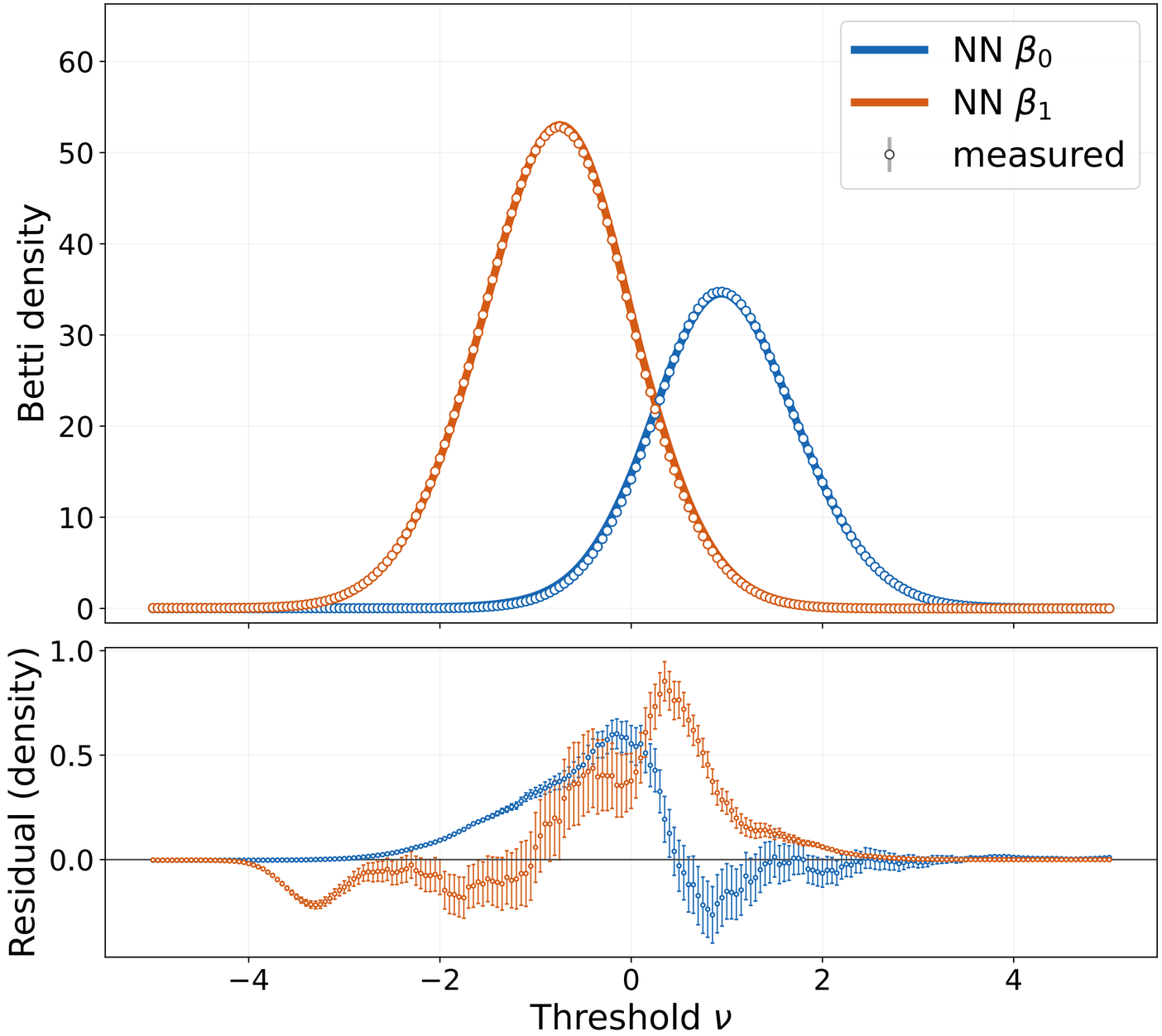
Inflation: $A_s(\frac{k}{k_*})^{n_s-1}e^{I_0(k)}[1+I_1^2(k)]$

Thermal WDM: $P_{\rm CDM}(k)[1+(\alpha k/h)^{2\mu}]^{-10/\mu}$

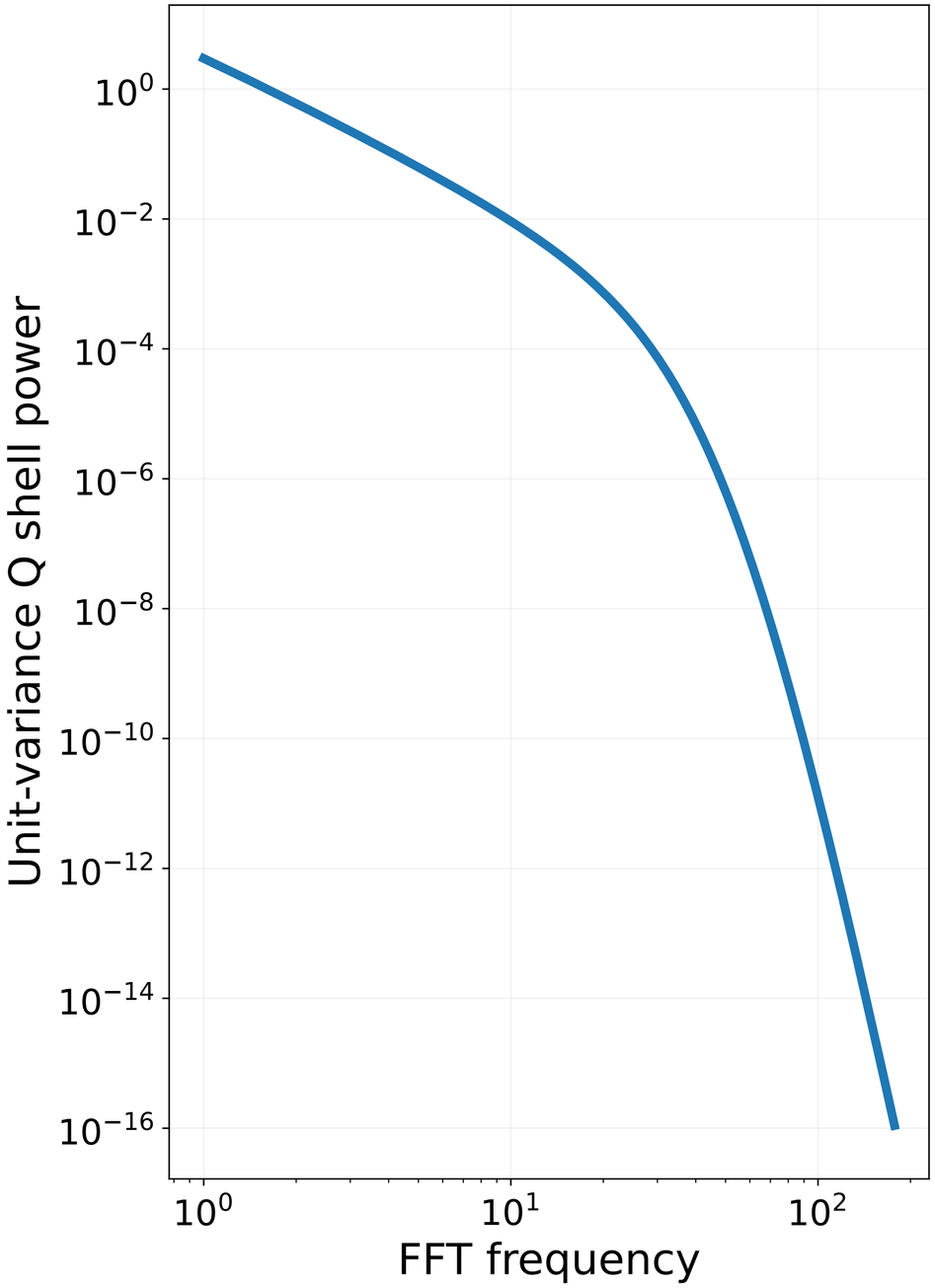
Evaluate out-of-distribution (OOD) generalization across different spectral families

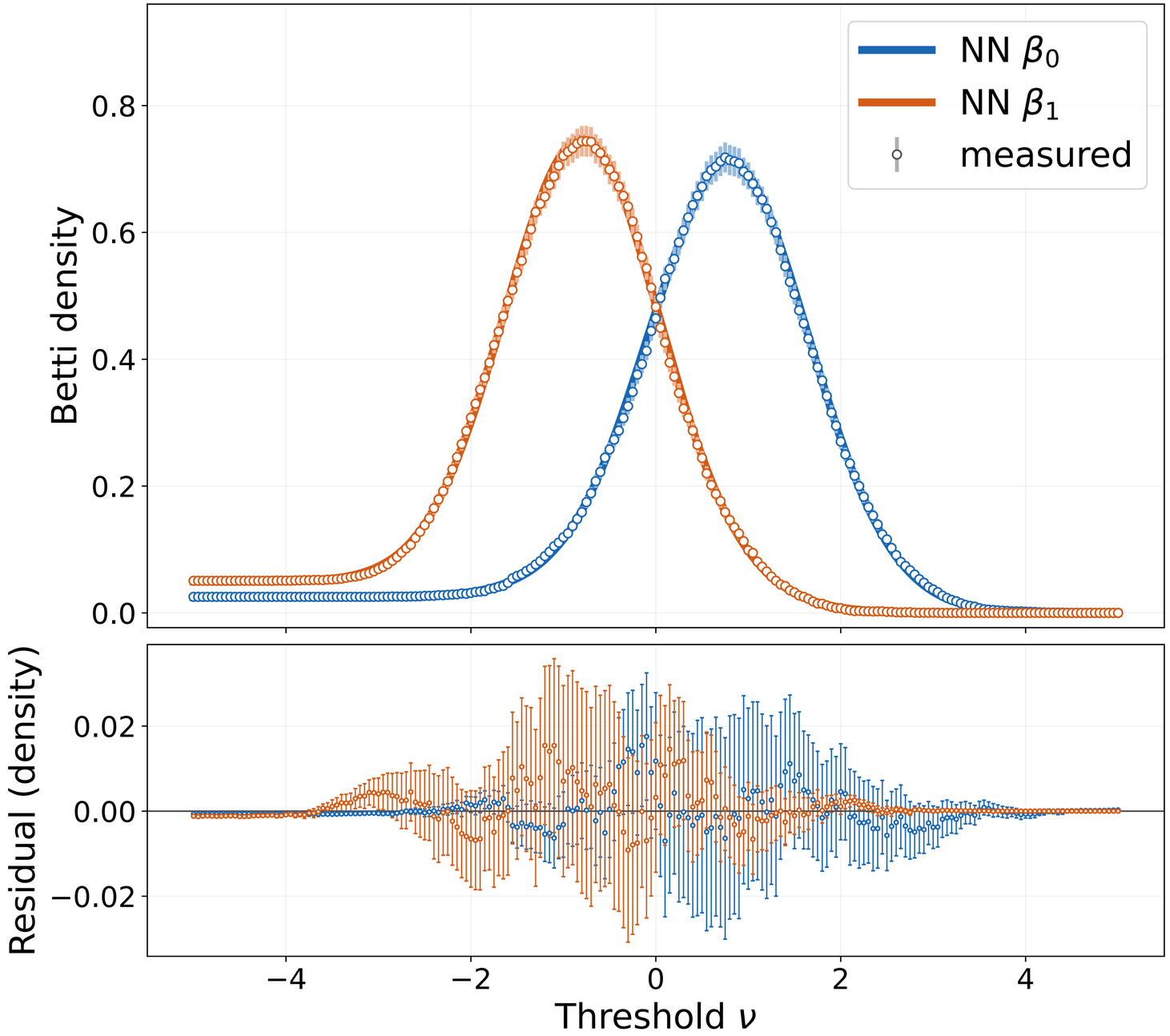
BettiAI v1 | Full 131K | Eisenstein-Hu BAO
Reused v8 bank | selected hard case 86 | ensemble L2 1.30%



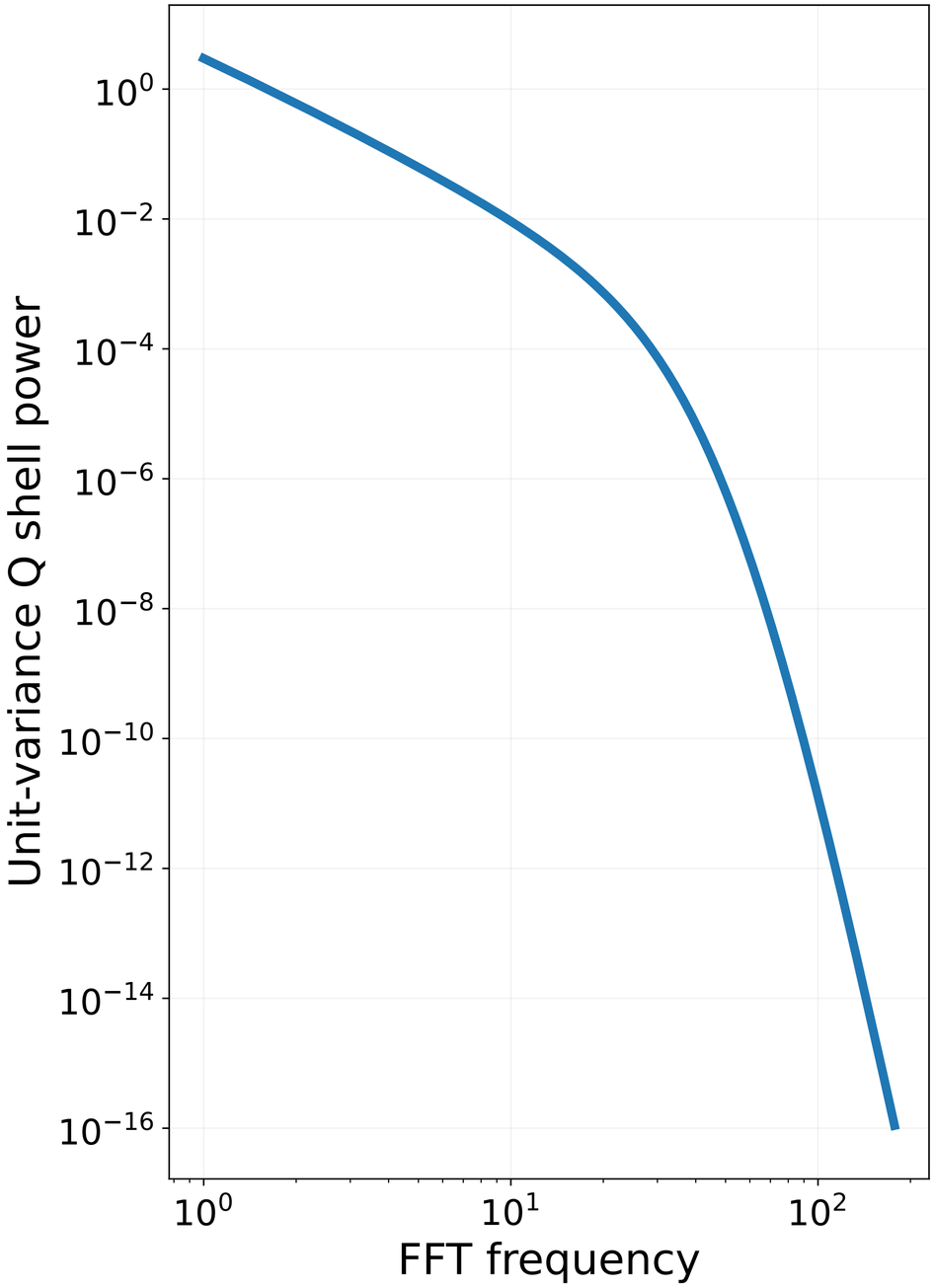


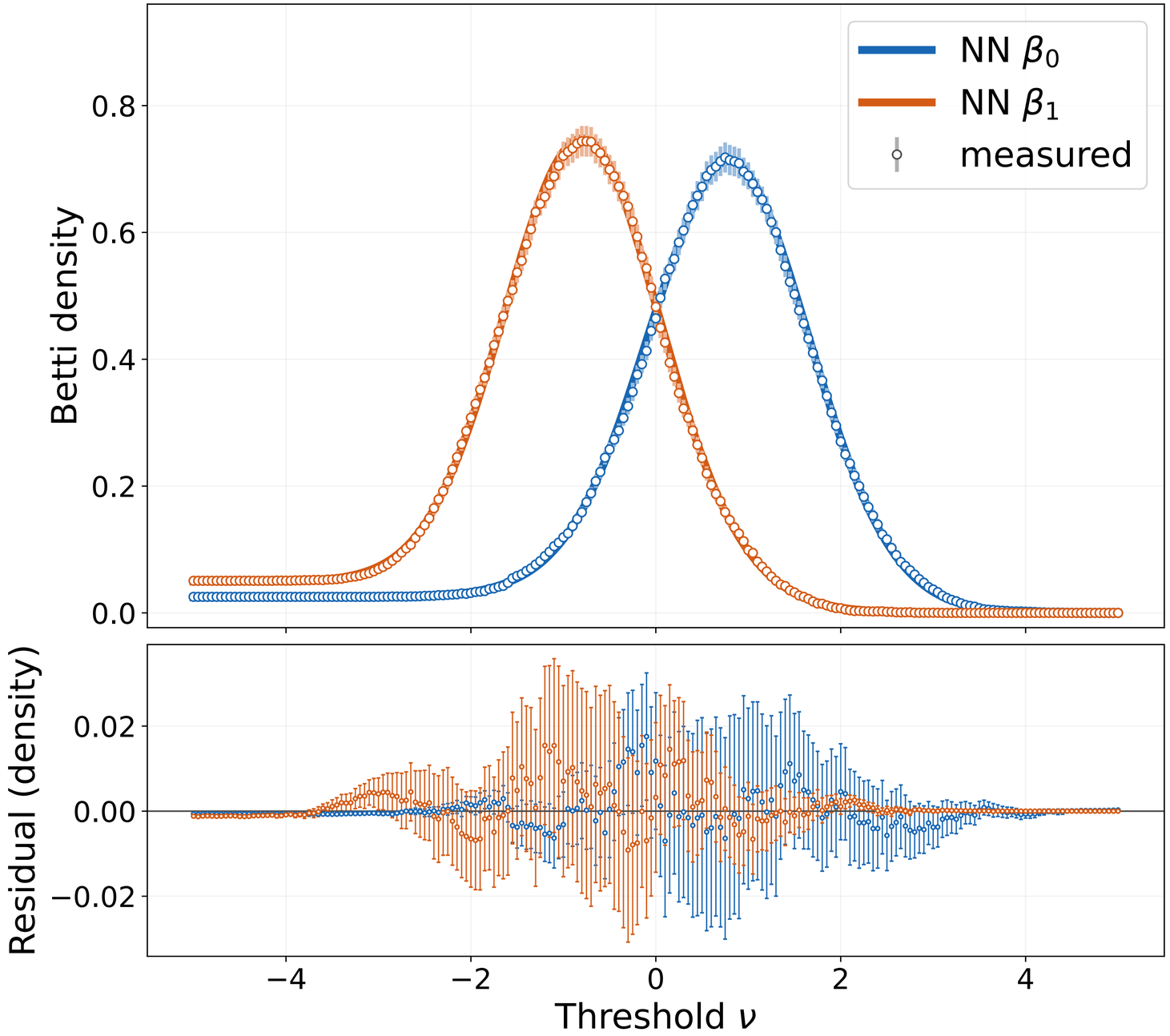
BettiAI v1 | Full 131K | Thermal WDM (feature in band)
Reused v8 bank | fixed case 64 | ensemble L2 1.38%





BettiAI v1 | Full 131K | Thermal WDM (feature in band)
Reused v8 bank | fixed case 64 | ensemble L2 1.38%





Summary

- **Neural operators** can predict $\langle \beta_i \rangle$ from $P(k)$ in Gaussian fields
  - **Challenge**: how to construct a training dataset $P(k)$ that cover the space of possible Betti curves? Non-Gaussian fields even more challenging

- **Further analytical reduction** provided by ChatGPT, implemented and evaluated symbolically in Mathematica, validated with numerical measurement
  - Extending this approach to predict other local geometric observables is a potential research direction

Thank you.